

37. Tubulointerstitial Diseases · assets 1-9 / 9

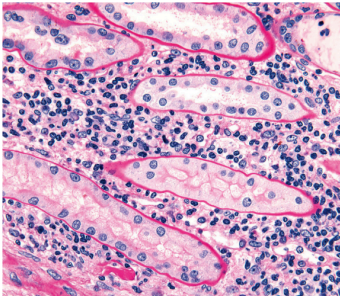


Fig. 37.1 Renal histology observed in acute tubulointerstitial nephritis highlighting the inflammatory infiltrate. Periodic acid–Schiff stain. (From Perazella MA. Clinical approach to diagnosing acute and chronic tubulointerstitial disease. *Adv Chronic Kidney Dis.* 2017;24:57–63.)

Fig_37_1

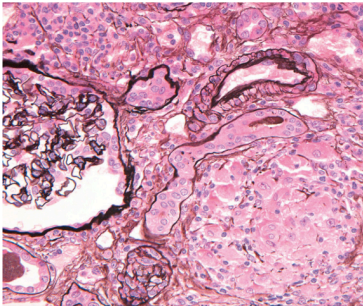


Fig. 37.2 Renal histology in sarcoidosis highlighting the inflammatory infiltrate and noncaseating granuloma. Jones silver stain. (From Perazella MA. Clinical approach to diagnosing acute and chronic tubulointerstitial disease. *Adv Chronic Kidney Dis*. 2017;24:57–63.)

Fig_37_2

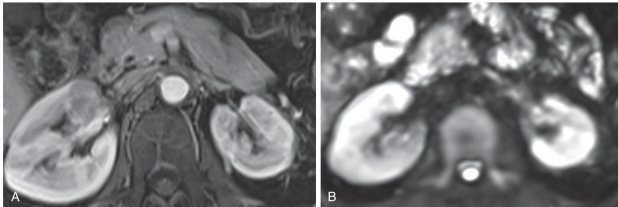


Fig 37.3 IgG4-related kidney disease emulating malignancy. (A) Magnetic resonance imaging (MRI) of the kidney with contrast showing multiple well-circumscribed right renal masses, the largest measuring $2.3 \times 1.9 \times 2.5$ cm, with imaging characteristics suggestive of a malignancy, such as papillary renal cell carcinoma or lymphoma. (B) MRI of the kidney shows resolution of the right renal-enhancing lesions after treatment with steroids. (Images courtesy Dr. Mohamed G. Atta.)

Fig_37_3

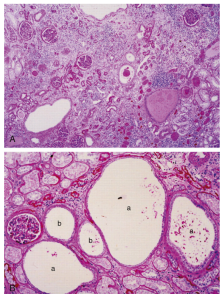


Fig. 37.4 (A) Severe lithium-associated chronic tubulointerstitial nephropathy, with the additional finding of focal tubular cysts arising in a background of severe interstitial fibrosis and tubular atrophy. (Periodic acid-Schiff [PAS] stain, $\times 40$) (B) High-power view of tubular cysts lined by simple cuboidal epithelium (a). Adjacent tubules show tubular dilation (b). (PAS stain, $\times 100$) [From Markowitz GS, Radhakrishnan J, Kamkhani N, et al. Lithium nephrotoxicity: a progressive combined glomerular and tubulointerstitial nephropathy. *J Am Soc Nephrol*. 2000;11:1439–1448.]

Fig_37_4

Table 37.1 Broad Causes of Tubulointerstitial Nephritis	
Broad Causes of Tubulointerstitial Nephritis	
Immune Mediated	Nonimmune Mediated
Cell Mediated	Genetics
Infection	Acute tubular necrosis
Infusion	subnephritic kidney diseases (ATN)
Kidney transplant rejection	Nephrotic syndromes (NPN)
	Cystinosis
	Karyomegaly: Interstitial Nephritis
	Nephritis
Autoimmune	Metabolic
Sjögren's syndrome	Hypertension
WGA-related disease	Cystitis (UTI, prostatic Ca, oxalate, urea)
Translational	Toxins
Tubulointerstitial nephritis with systemic syndrome	Hemolytic
Acute tubulointerstitial nephritis	Lithium
Acute tubulointerstitial nephritis	Heavy metals
acute (APN)	Cystitis
Antibiotic toxicity: antibiotics	Radiation
	Angiogenesis
	Endocrine neoplasia (Balkan, OGDs)
	Others
	Respiratory
	Neurological
	Neurodegenerative
	diseases, plasma cell dyscrasias
	Lymphoma
Examples of Systemic Diseases Associated with Secondary Tubulointerstitial Nephritis	
Systemic lupus erythematosus	Diabetes mellitus
Sarcoidosis	Cystic kidney diseases
Alcoholism	Acute tubular necrosis
Vasculitis	Acute tubular necrosis

Table_37_1

Class	Specific Causes of Acute Toxicosis
Glucose	Specific Glucose Alkalosis
Dog 10%	
Antiemetics	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
MAOIs and COOH-D inhibitors	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Drugs	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Anticancer agents	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Miscellaneous	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Systemic Diseases 20%	
Infection	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Infection Agents 6%	
Bacteria	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Fungi	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Others	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Toxicologic	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)
Therapeutic	1-anticholinergics (atropine, scopolamine) 2-phenothiazines (acepromazine, acepromazine, acepromazine) 3-metoclopramide (Reglan) 4-marijuana (marijuana)

Table_37_2

Author, Year	Design	N	Etiology	Arms	Storage Regimen	Time to Steroid Treatment	Results
Clarkson et al., 2004 ¹	Retrospective Single center 12 mo follow-up	42	82% Drugs 6% Idiopathic	16 Steroids 16 No Steroids Consentment	N Methylprednisolone 500 mg/day tapered 2.4 mg/day Or prednisolone 0.75 mg/kg/day tapered over 3-6 weeks	3.5 Weeks (mean time from symptom onset to biopsy)	No differences between groups in median survival or clearance at 1 year
Gonzalez et al., 2008 ²	Retrospective Multicenter 19 mo follow-up	61	100% Drugs	85% Steroids 16% No Steroids	N Methylprednisolone 250-500 mg/day tapered over 8-12 weeks Or prednisolone 1 mg/kg/day tapered over 8-12 weeks	13 to 24 days in those with complete recovery 16 to 24 days in those with incomplete recovery (p = 0.02)	54% Complete recovery
Rizza et al., 2012 ³	Retrospective Single center 19 mo follow-up	49	67% Drugs Other: 4 THN 2 Sarcoidosis 10 idiopathic	75% Steroids Other	Or prednisolone 1 mg/kg/day	2.8 days (mean time to hospital admission)	Greater improvement in eGFR (4.4 vs. 2.1 mL/min/1.73 m ² ; P = 0.05) Prolonged need for dialysis (18 vs. 42%; P = 0.005)
Muzilli et al., 2014 ⁴	Retrospective Single center 6-mo follow-up	133	71% Drugs 29% Other	86% Steroids 14% No Steroids	21% initially received IV steroids Starting dose of oral steroids was 60-80 mg/day with a median treatment duration of 7.1 weeks	8 vs. 11 vs. 35 days in those with complete, partial and no recovery respectively (P = 0.05)	49% CR 32% PR
Valuel et al., 2015 ⁵	Retrospective Multicenter 12 mo follow-up	171	73% Drugs 10% Sarcoid/TN 6% Immune 8% Infection 7% Unknown	63% Steroids 37% No Steroids	Unknown	41.1% Early (at 7 days) 48% Late (7 days) 11% unknown	No difference in CR (48% vs. 41%) Dialysis dependence was 10% in both groups
Ramachandran, et al., 2015 ⁶	Prospective randomized Single center 12 week follow-up	29	100% Drugs	100% Steroids	Group 1: oral prednisolone 1 mg/kg/day for 3 weeks, followed by rapid tapering in the next 3 weeks Group 2: IV methylprednisolone 30 mg/kg/day max 1 g for 3 days followed by oral prednisolone 1 mg/kg/day for 2 weeks and tapered in next 3 weeks	1 Week after withdrawal of corticoid drug in all cases	Oral and pulse steroid therapy equally effective in achieving remission

Table_37_3_part1

Author, Year	Design	N	Etiology	Arms	Steroid Regimen	Time to Steroid Treatment	Results
Predecki et al, 2017 ⁶¹	Retrospective Single center 39-mo follow-up	187	25% Drugs 48% Unknown 13% TB 9% Sarcoid 3% TINU 2% Sjögren	84% Steroids	Steroid therapy was determined by clinician choice. All received oral Prednisolone at doses of 40–60 mg/day Only 3 patients received methylprednisolone before oral prednisolone Group A: oral prednisolone 1 mg/kg/day for 2 weeks Group B: pulse methylprednisolone 30 mg/kg/day for 3 days (maximum 1 g), followed by oral prednisolone 1 mg/kg/day for 2 weeks, tapered over 2 weeks Initial dose of prednisolone: 0.8 ± 0.2 mg/kg/day for 2 weeks	NR	Steroid-treated patients had better eGFR at 2 years (43 vs. 24 mL/min) and less dialysis dependence (5.1% vs. 24.1%)
Chowdry et al, 2018 ⁶²	Prospective randomized 3-mo follow-up	31	100% Drugs	100% Steroids	Group A: oral prednisolone 1 mg/kg/day for 2 weeks Group B: pulse methylprednisolone 30 mg/kg/day for 3 days (maximum 1 g), followed by oral prednisolone 1 mg/kg/day for 2 weeks, tapered over 2 weeks 48% IV steroids before oral prednisolone	1 Week after withdrawal of causative drug	58% CR 42% PR There was no significant difference between the 2 groups
Fernandez-Juanes et al, 2018 ⁶³	Retrospective Multicenter 6-mo follow-up	182	100% Drugs	100% Steroids	48% IV steroids before oral prednisolone Initial dose of prednisolone: 0.8 ± 0.2 mg/kg/day for 2 weeks	12 Days (mean interval between diagnosis and steroid initiation)	41% CR 46% PR
Surendra et al, 2019 ⁶⁴	Retrospective Single center 12-mo follow-up	83	100% Drugs	100% Steroids	500 mg Methylprednisolone for 3 days, followed by prednisolone of 0.5–1 mg/kg/day tapered over 4–6 weeks	5–7 days from drug withdrawal	47% CR 53% PR

CR, Complete recovery; D-ATIN, drug-induced acute tubulointerstitial nephritis; eGFR, estimated glomerular filtration rate; NR, not reported; PR, partial recovery; SJc, Sjögren syndrome.

Table_37_3_part2

Table 37.4 Causes of Chronic Tubulointerstitial Nephritis	
Disease	Specific Causative Agents/Disease Mutations
Genetics	
Autosomal dominant tubulointerstitial kidney disease (ADTKD)	ADTKD-JAKOJ: Uromodulin (AKOJ) gene mutations in chromosome 16p12.3 ADTKD-MKOJ: Mucin 1 (MKOJ) mutations in chromosome 10p21 ADTKD-HEN: Renin (REN) mutations in chromosome 12q13.1 ADTKD-HNF1B: Hepatocyte nuclear factor 1B (HNF1B) mutations in chromosome 17p12 ADTKD-BESGAT1: SESO3 mutations in chromosome 3q21.3 CTNS gene mutations in chromosome 17p13
Cystinosis	Fanconi anemia-associated uroliths (FAU) gene mutations
Acrolymphagial interstitial nephritis (APIN)	Autosomal recessive mutations in the (FAPN1-20 genes) that encode nephrins/protins
	APN1 is the most frequent
	CHAB11, P302 mutations
Others	
Toxic Factors and Potential Sources of Exposure	
Endemic nephritis	Isti-a-tia disease (Apari): Linked to Chinese herb exposure in water, copper, mining, pollution Balkan endemic nephropathy or Chinese herb nephropathy: Caused by aristolochic acid-containing herb OKD Mesonephros/nephros Sri Lanka OKD India OKD
Heavy metal toxicity	Arsenic: Groundwater, manufacturing (pylwood, others) Chromium: Mining, environmental lead Cadmium: Mining, electrolytic Lead: Mining, refining, paint, battery manufacturing, monomers Mercury: Manufacturing (batteries, pig, latex), cosmetics, traditional medicines Thallium: Mining; contaminated groundwater; warfare Toxicity of neurotoxic/chemotoxic, nephrotoxic, hepatotoxic, oxidant, anti acid In cancer treatments, occupational, nuclear plant accidents, warfare
Lithium	
Medicative	
Radiation exposure	
Others	
Plants, mushrooms, and herbal supplements	Corticosteroids and/or Opioid/steroids or, Anandamide synthase mushrooms, Aristolochic acid-containing plants, and, Medications
Vaccinocellular reflux	
Lymphoproliferative disorders, plasma cell dyscrasias	
Others	
Chronic kidney disease of unknown etiology	

Table_37_4